Event Boundaries and Movie Memory

BRSM Report 1: Team Odomos

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March 10, 2026

Background: Event Segmentation Theory

- Our brains **segment** continuous experience into discrete *events*
- Transitions = **event boundaries** (Zacks et al., 2007)
- Boundaries trigger an update of the internal *event model*
- Memory at event boundaries is encoded **differently** (Swallow et al., 2009)

Two types of boundaries:

- **Natural boundaries (NB):** Smooth, organic transitions
- **Abrupt boundaries (AB):** Sudden, artificially imposed transitions (hard cuts)

Task Paradigm

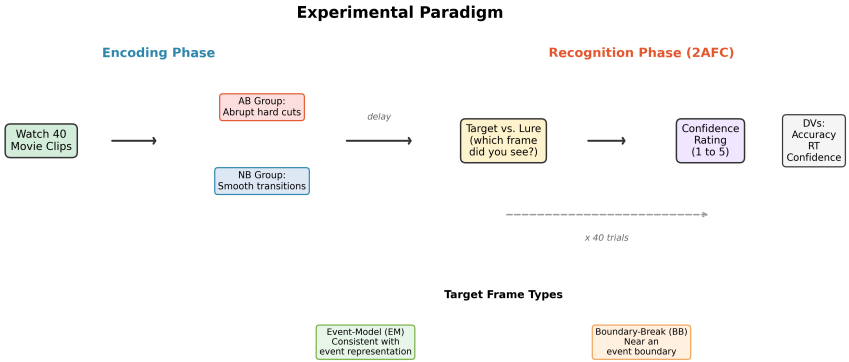


Figure 1: Experimental paradigm: encoding phase (AB vs. NB movie clips), followed by 2AFC recognition test with confidence ratings.

Design and Participants

- **2 $\times$ 2 mixed design:**
  - Boundary Type (AB vs. NB) – **between-subjects**
  - Target Frame Type (EM vs. BB) – **within-subjects**
- **170 participants** (81 AB, 89 NB); 1 excluded for missing data
- 40 trials per participant (20 EM, 20 BB) = **6,800 trials total**
- **DVs:** Recognition accuracy, Response time, Confidence (1–5)

Hypotheses

H1 – Boundary Type main effect:

- NB group will show better recognition performance than AB group

H2 – Target Type main effect:

- EM targets will be recognized more accurately and confidently than BB targets

H3 – Interaction:

- The EM–BB gap may be larger in the AB group (abrupt boundaries differentially impair boundary-adjacent content)

Each hypothesis tested for all three DVs: Accuracy, RT, Confidence.

Data Processing and Analytical Pipeline

- **171 PsychoPy CSV files** parsed in Python 3
  - 1 excluded (sub42); Final: **170 participants, 6,800 trials**
- Per trial: accuracy (0/1), RT (s), confidence (1–5)
- RT outliers removed: < 0.2 s or > 60 s (1 trial)

For each DV, the analysis pipeline was:

1. Descriptive statistics (means, SDs)
2. **Normality check** (Shapiro-Wilk per cell, Levene's test)
3. 2×2 mixed ANOVA (η_p^2; follow-ups with Cohen's *d*)
4. **If normality violated** $\rightarrow$ non-parametric robustness checks
   - Mann-Whitney U (between), Wilcoxon (within), Mann-Whitney on diffs (interaction)

H1, H2, H3: Accuracy – Descriptive Statistics

Condition	Target	Accuracy ($M \pm SD$)
AB	EM	$.855 \pm .092$
AB	BB	$.824 \pm .105$
NB	EM	$.883 \pm .079$
NB	BB	$.860 \pm .096$

- All well above chance (50%)
- **H1 pattern:** NB ($M = .871$) $>$ AB ($M = .840$) (consistent)
- **H2 pattern:** EM ($M = .870$) $>$ BB ($M = .843$) (consistent)

Accuracy – Normality Check and Inferential Tests

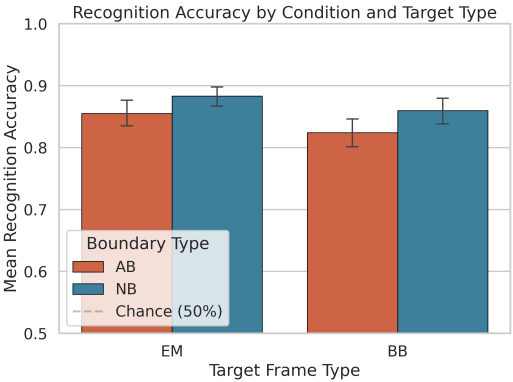
Normality: Violated in all 4 cells ($p < .002$)
Levene's: variance homogeneity OK ($p > .08$)

Parametric (mixed ANOVA):

- H1: $F(1, 168) = 7.25$, $p = .008$, $\eta_p^2 = .041$ (supported)
- H2: $F(1, 168) = 11.44$, $p < .001$, $\eta_p^2 = .064$ (supported)
- H3: $F(1, 168) = 0.21$, $p = .651$ (n.s.)

Non-parametric (normality violated):

- H1: Mann-Whitney $U = 2744$, $p = .007$ (supported)
- H2: Wilcoxon $W = 3135$, $p = .003$ (supported)
- H3: $U = 3644$, $p = .904$ (n.s.)



Accuracy – Effect Sizes

- **H1: NB > AB**, $d = 0.41$ (medium effect)
  - NB ($M = .871$) vs. AB ($M = .840$)
- **H2: EM > BB**, $d = 0.29$ (small–medium effect)
  - EM ($M = .870$) vs. BB ($M = .843$)
- **H3: No interaction** ($p = .651$)
  - The EM–BB advantage is similar in both groups

→ Natural boundaries help memory; EM frames help memory; effects are additive.

Now Hrishiraj will present the RT and confidence results.

H1, H2, H3: Response Time

Normality: Violated in all 4 cells ($p < .002$)
Levene's: OK ($p > .08$)

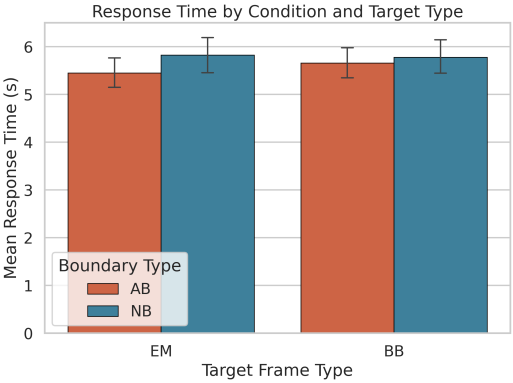
Descriptive: RT = 5.45 – 5.83 s; small differences.

Parametric ANOVA:

- H1: $F(1, 168) = 1.10$, $p = .296$ – n.s.
- H2: $F(1, 168) = 1.16$, $p = .284$ – n.s.
- H3: $F(1, 168) = 3.36$, $p = .069$, $\eta_p^2 = .020$

Non-parametric: All n.s. ($p > .09$)

Near-significant interaction warrants future investigation.



H1, H2, H3: Confidence – Descriptive + Normality

Condition	Target	Confidence ($M \pm SD$)
AB	EM	$4.13 \pm .46$
AB	BB	$4.04 \pm .49$
NB	EM	$4.21 \pm .44$
NB	BB	$4.20 \pm .42$

- **H2 pattern:** EM $>$ BB within AB, but **not within NB** (possible interaction)
- **Normality:** 3 cells normal; NB $\times$ EM violated ($p = .004$)
- Levene's: variance homogeneity OK

H1, H2, H3: Confidence – Inferential Tests

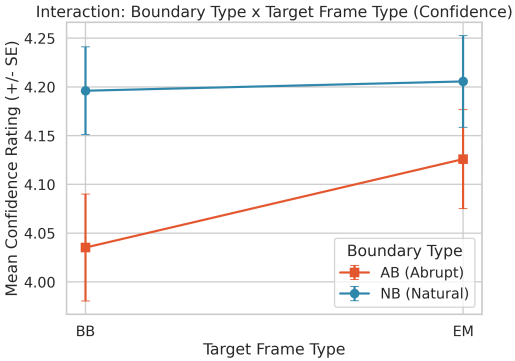
Parametric ANOVA:

- H1: $F(1, 168) = 3.24$, $p = .074$ – n.s.
- H2: $F(1, 168) = 5.70$, $p = .018$, $\eta_p^2 = .033$ (supported)
- **H3: $F(1, 168) = 4.03$, $p = .046$, $\eta_p^2 = .023$** (supported)

Simple effects (H3):

- AB: EM > BB ($p = .003$, $d = .19$)
- NB: EM ≈ BB ($p = .733$) – no difference
- BB: NB > AB ($p = .024$, $d = .35$)

Non-parametric: Wilcoxon confirmed H2 ($p = .025$). Mann-Whitney confirmed H3 ($p = .025$).



Key Finding: Accuracy vs. Confidence Dissociation

	H1 (Boundary)	H2 (Target)	H3 (Interaction)
Accuracy	Significant	Significant	**Not significant**
Confidence	Not significant	Significant	**Significant**

- Accuracy: both main effects, **no interaction** (effects are additive)
- Confidence: **significant interaction** (effects are not additive)
- AB group is less confident about BB targets, but NB group is equally confident
- Abrupt boundaries **selectively impair metacognitive certainty** for boundary-adjacent content

Correlations and Summary

Spearman correlations (between DVs):

- Accuracy–Confidence: $\rho = 0.360$, $p < .0001$ (metacognitive calibration)
- RT–Confidence: $\rho = -0.152$, $p = .047$ (faster = more confident)
- Accuracy–RT: $\rho = -0.054$, $p = .487$ (no speed–accuracy tradeoff)

Summary of hypothesis tests:

- **H1 supported** for accuracy ($d = 0.41$); NB > AB
- **H2 supported** for accuracy ($d = 0.29$) and confidence ($d = 0.11$); EM > BB
- **H3 supported** for confidence only: AB group less confident about BB targets
- All conclusions robust to non-parametric testing

Limitations and Future Directions

- No demographic data captured (no individual difference analyses)
- Item-level variability not modelled
  - Future: mixed-effects models with crossed random effects
- Signal Detection Theory (d', criterion c) could add nuance
- Near-significant RT interaction ($p = .069$) warrants follow-up
- **All non-parametric tests confirmed parametric conclusions**

Thank You

Team Odomos

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Questions?

References

- Swallow, K. M., Zacks, J. M., & Abrams, R. A. (2009). Event boundaries in perception affect memory encoding and updating. *Journal of Experimental Psychology: General*, *138*(2), 236–257.
- Zacks, J. M., & Swallow, K. M. (2007). Event segmentation. *Current Directions in Psychological Science*, *16*(2), 80–84.
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